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Semiconductor module

Abstract

A semiconductor module includes a semiconductor element and a molded resin case having a sidewall portion surrounding the semiconductor element. The sidewall portion has an upper surface to which a printed board is to be attached. The sidewall portion has an attachment hole from the upper surface thereof in a depth direction orthogonal to the upper surface, through which a screw is inserted to attach the printed board. The attachment hole is integrally provided in the case. The attachment hole has a columnar shape as a whole, and the attachment hole has a tip portion having a hemispherical shape.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This is a continuation application of International Application PCT/JP2020/024723 filed on Jun. 24, 2020 which claims priority from a Japanese Patent Application No. 2019-144576 filed on Aug. 6, 2019, the contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

Technical Field

(1) The present invention relates to a semiconductor module including a semiconductor element.

Background Art

(2) A semiconductor module has a substrate having thereon a semiconductor element such as an insulated gate bipolar transistor (IGBT), a power metal oxide semiconductor field effect transistor (MOSFET) and a free wheeling diode (FWD) and is utilized in, for example, an inverter apparatus.

(3) Such a semiconductor module includes a case for attaching a circuit substrate. The case is molded by resin molding such as injection molding. A melted resin is filled in a cavity space formed inside a die, and the resin is hardened to form a molded product. Attachment means for attaching a circuit substrate to the case of a semiconductor apparatus described above has been known in which screw seats of metallic columnar blocks are embedded (see Patent Literature 1, for example).

Prior Art Document

Patent Document

(4) [Patent Document 1]

(5) Japanese Patent Laid-Open No. 2001-36003

SUMMARY OF THE INVENTION

(6) However, the case according to Patent Literature 1 requires metallic screw seats separately from the case, requiring part costs and time and labor for attaching the screw seats. As shown in FIG. 5, when an attachment hole **70** for attaching a printed board **4** with a tapping screw **18** is resin-molded integrally with a case **50**, a corner **70b** is formed at an end part **70a** of the attachment hole. Thus, when the attachment hole **70** is formed integrally with the case **50** during resin molding of the case **50**, the resin may be retained around the end part **70a** of the attachment hole **70**. Furthermore, when the resin molding is performed, an outer side of the case **50** is cooled first and an inner side thereof is cooled later. As a result, the case **50** is pulled because of the differential shrinkage between the outer side and inner side of the case **50**. Therefore, a crack **81** or a vacant space (void) **82** may occur in a detail within the case **50**.

(7) When a crack **81** or a vacant space **82** occurs near the attachment hole **70** and if tapping is performed on the attachment hole **70** with the tapping screw **18**, an insufficient strength occurs, and there is a possibility that the case is damaged when the tapping screw **18** is fastened. When a crack **81** or a vacant space **82** occurs near the attachment hole **70** and if tapping is performed on the attachment hole **70** with the tapping screw **18**, there is a risk that a thread that is threaded decays with the tapping, torque loss occurs before a predetermined torque is acquired, and the tapping screw **18** turns free.

(8) The present invention has been made in view of such a point, and it is one of objects of the present invention to provide a semiconductor module in which occurrence of a crack or a vacant space (void) during resin-molding of a case of the semiconductor module can be deterred.

(9) According to one aspect, in a semiconductor module according to an embodiment including a semiconductor element, the semiconductor module has a case accommodating the semiconductor element, an attachment hole for attaching a printed board to an upper surface of the case with a screw is provided integrally with the case by resin molding, and the attachment hole has a

columnar shape as a whole, and the attachment hole has a tip having a hemispherical shape.

(10) According to the present invention, a semiconductor module can be provided in which occurrence of a crack and a vacant space (void) during resin molding of a case of a semiconductor apparatus can be deterred.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic cross-sectional view showing an example of a semiconductor apparatus including a semiconductor module according to an embodiment.
- (2) FIG. 2 is a plan view showing an example of a case of the semiconductor module according to the embodiment.
- (3) FIG. 3 is a cross-sectional view of the case shown in FIG. 2 taken at a line a-a' in a thickness direction viewed from a direction indicated by arrows.
- (4) FIG. 4 shows an enlarged view of the cross-sectional view of an attachment hole shown in FIG. 3.
- (5) FIG. 5 shows an example of a conventional resin-molded case.

DESCRIPTION OF EMBODIMENTS

- (6) A semiconductor apparatus to which the present invention is applicable is described below. FIG. 1 is a schematic cross-sectional view showing an example of a semiconductor apparatus **100** including a semiconductor module **1** according to an embodiment. The semiconductor apparatus to be described below is merely an example and is not limited thereto but can be changed as required. The expression “planar view” herein refers to a view of the semiconductor apparatus from a direction perpendicular to an insulating circuit substrate.
- (7) The semiconductor apparatus **100** is applied to a power converter such as a power module. As shown in FIG. 1, an insulating circuit substrate **2** is attached to the semiconductor module **1**. The semiconductor module **1** includes a base plate **12** being a heat dissipation plate, a semiconductor element **3** disposed on an upper surface of the insulating circuit substrate **2**, and a case **13** that surrounds a circumference of the insulating circuit substrate **2** and the semiconductor element **3**. The insulating circuit substrate **2** is disposed on an upper surface of the base plate **12**. The semiconductor module **1** is attached to an external apparatus **11** such as a cooler. A case **13**, which constitutes a semiconductor module, has a printed board attached to the upper surface of the case **13**. The printed board is described below. The case **13** has attachment holes for attaching the printed board with tapping screws. The attachment holes are explained with FIGS. 2-4 later. The tapping screws are explained with FIGS. 3 and 4 later.
- (8) The external apparatus **11** is configured by a cooler such as a heat sink, heat dissipation fins, and a water cooling jacket and has a square shape in planar view. The external apparatus **11** has an upper surface having a smooth bonding surface to which the semiconductor module **1** is bonded. The external apparatus **11** is formed of metal such as copper or aluminum or an alloy containing one kind or more of them, and, for example, plating processing is performed on the surface of the external apparatus **11**. Having described the example in which, according to the embodiment, the external apparatus **11** is configured by a cooler, embodiments are not limited thereto. For example, the external apparatus **11** may be configured by a unit that is not shown.
- (9) The base plate **12** functions as a heat dissipation plate, is formed by performing plating processing on a surface of a metallic plate of, for example, copper, and has, for example, a rectangular shape in planar view. The base plate **12** has an upper surface having a smooth bonding surface to which the insulating circuit substrate **2** is bonded.
- (10) The insulating circuit substrate **2** is formed by stacking a metallic layer and an insulating layer and has a rectangular shape in planar view smaller than the upper surface of the base plate **12**.

More specifically, the insulating circuit substrate **2** has an insulating plate **20** having an upper surface (one surface) and a lower surface (another surface) on the opposite side of the upper surface, a first metallic layer **21** provided on the upper surface of the insulating plate **20**, and a second metallic layer **22** provided on the lower surface of the insulating plate **20**. The thicknesses of the insulating plate **20**, the first metallic layer **21** and the second metallic layer **22** may be equal or may be different.

(11) The insulating plate **20** is formed by an insulator of, for example, ceramic, and the first metallic layer **21** and the second metallic layer **22** are formed of, for example, a copper foil. The first metallic layer **21** configures a circuit layer electrically connected to a semiconductor element. The first metallic layer **21** has a flat surface and is formed by disposing a predetermined circuit pattern on the upper surface of the insulating plate **20**. More specifically, an outer edge part of the first metallic layer **21** is positioned slightly inner than an outer edge part of the insulating plate **20**.

(12) The second metallic layer **22** has a flat surface and has a rectangular shape in planar view that covers the substantially entire lower surface of the insulating plate **20**. More specifically, an outer edge part of the second metallic layer **22** is positioned slightly inner than the outer edge part of the insulating plate **20**. The lower surface of the second metallic layer **22** of the insulating circuit substrate **2** is bonded to the upper surface of the base plate **12**.

(13) The insulating circuit substrate **2** configured as described above is formed by, for example, a direct copper bonding (DCB) substrate or an active metal brazing (AMB) substrate. The insulating plate **20** is formed by, for example, a ceramics material such as alumina (Al.sub.2O.sub.3), aluminum nitride (AlN) or silicon nitride (Si.sub.3N.sub.4). The insulating circuit substrate **2** is disposed at, for example, a center part of the upper surface of the base plate **12**. [0019]

(14) A semiconductor element **3** is disposed on the upper surface of the first metallic layer **21** of the insulating circuit substrate **2**. The semiconductor element **3** is formed by a semiconductor substrate of, for example, silicon (Si) or silicon carbide (SiC). The semiconductor element **3** has, for example, a rectangular (square) shape in planar view. Three semiconductor elements **3** are disposed at a substantial center part on the first metallic layer **21**. All of the semiconductor elements **3** are disposed on the first metallic layer **21** through a bonding material such as solder S. Thus, the semiconductor elements **3** are electrically connected to the first metallic layer **21**. The number and positions of the semiconductor elements **3** are not limited thereto but can be changed as required.

(15) As the semiconductor element **3**, a switching element such as an insulated gate bipolar transistor (IGBT) and a power metal oxide semiconductor field effect transistor (MOSFET), or a diode such as a free wheeling diode (FWD) is used. Alternatively, as the semiconductor element **3**, a reverse conducting (RC)—IGBT in which an IGBT and an FWD are integrated, a reverse blocking (RB)—IGBT having a sufficient resistance to pressure against a reverse bias or the like may be used.

(16) The case **13** is a housing that accommodates the insulating circuit substrate **2** and the semiconductor elements **3**. The case **13** is molded by a synthetic resin molded by injection molding and surrounds an outer circumferential side of the insulating circuit substrate **2**. The case **13** and a lid part, which is described below, are formed of a thermoplastic resin. As such a resin, a PPS resin, a PBT resin, a PBS resin, a PA resin, an ABS resin or the like is available. A filler may be added to those resins as required. The case **13** has a lower surface that faces the upper surface of the base plate **12** and is bonded to the insulating circuit substrate **2** with a bonding material (adhesive) not shown.

(17) The case **13** is disposed around the insulating circuit substrate **2** and the semiconductor element **3** and defines a space in which a sealing resin **15**, which is described below, is filled. The case **13** has a square ring shape in planar view corresponding to the external shape of the base plate **12** and projects in a thickness direction (vertical direction) of the insulating circuit substrate **2**. External terminals **14** are embedded in the sides that define the square ring shape of the case **13**. Each of the external terminals **14** has an L shape in cross-sectional view and has one end projecting

from an inner wall surface of the case **13** and another end projecting from the upper surface of the case **13**. The external terminals **14** are integrated to the case **13** by, for example, insert molding. Details of the case **13** are described below.

(18) The three semiconductor elements **3** are electrically connected by a wiring member **W**. Partial semiconductor elements **3** are electrically connected to the external terminals **14** on one side through a wiring member **W** described above. The external terminals **14** on the other hand are electrically connected to the first metallic layer **21** through a wiring member **W**.

(19) A conductive wire is used for the wiring members **W**. The material of the conductive wire can be one or a combination of gold, copper, aluminum, a gold alloy, a copper alloy and an aluminum alloy. A member excluding the conductive wire can be used as the wiring members. For example, a ribbon or a lead frame can be used as the wiring members.

(20) An internal space of the case **13**, which is defined by the case **13**, is sealed by the sealing resin **15**. More specifically, the sealing resin **15** partially seals the insulating circuit substrate **2**, the semiconductor elements **3**, the wiring members **W** and the external terminals **14** in the inner side of the case **13**. As the sealing resin **15**, an epoxy resin or a silicone gel can be used. The sealing resin **15** is filled within the case **13** up to the height where the wiring members **W** described above are buried. After the sealing resin **15** is filled, a lid part **16** is attached above the sealing resin **15**. The lid part **16** protects the inside of the case **13** by covering the upper part of the insulating circuit substrate **2** and the semiconductor elements **3**.

(21) FIG. **2** is a plan view showing an example of the case **13** of the semiconductor module **1** according to the embodiment. FIG. **3** is a cross-sectional view of the case shown in FIG. **2** taken at a line a-a' in a thickness direction viewed from a direction indicated by arrows. FIG. **4** shows an enlarged view of the cross-sectional view of an attachment hole shown in FIG. **3**.

(22) Attachment holes **17a**, **17b**, **17c** and **17d** for attaching a printed board **4** with screws are provided in vicinity of corners of an outer edge of the case **13**. The attachment holes **17a**, **17b**, **17c** and **17d** are provided at positions corresponding to through-holes **28** of the printed board **4**. Hereinafter, the attachment holes **17a**, **17b**, **17c** and **17d** are called "attachment holes **17**" when no particular distinction is required among them. The attachment holes **17** are integrated to the case **13**.

(23) The attachment holes **17** are provided as thread holes for tapping screws **18**. The tapping screws **18** are fastened by the attachment holes **17** through the through-holes **28** of the printed board **4** so that the printed board **4** is fixed onto the case **13**. Each of the attachment holes **17** has a columnar shape as a whole and has a circular shape in planar view. The column includes not only a shape having a bottom surface being a perfect circle but also a slightly tilted truncated cone. The inner part of the attachment hole **17** is not threaded and does not have a thread before the printed board **4** is attached with the tapping screws **18**. A thread is not formed until the inner part of the attachment hole **17** is threaded by the tapping screw **18** to attach the printed board **4** with the tapping screws **18**. Each of the attachment holes **17** has a diameter **d1** equal to or larger than 1.0 mm and equal to or smaller than 5.0 mm. Preferably, the diameter **d1** is equal to or larger than 2.0 mm and equal to or smaller than 3.0 mm.

(24) An end part **19** of each of the attachment holes **17** has a hemispherical shape. Preferably, the end part **19** has an R shape corresponding to a radius of the hole diameter. The hemispherical shape includes not only a shape cut at a plane extending through the center of a sphere but also a shape cut at a plane not extending through the center of a sphere. The sphere includes not only a shape of a perfect circle but also a shape of an ellipsoid. The expression "R shape corresponding to a radius of the hole diameter" refers to a space shape formed by a dimension of a radius **r** of the attachment hole **17** at each of the end parts **19** of the attachment holes **17**. The dimension of the radius **r** of the attachment hole **17** is calculated based on the diameter **d1** of the attachment hole **17**. Thus, because a resin material flows smoothly along the rounded end parts **19** during resin molding even when the attachment holes **17** are integrally molded with the case **13**, occurrence of a crack (see FIG. **5**) and

a vacant space (void) **82** (see FIG. 5) within the case **13** can be deterred. Furthermore, the rounded end part **19** can prevent concentration of stress to the end part **19**, and occurrence of a crack **81** and a vacant space **82** within the case **13** can be deterred.

(25) Also, occurrence of a crack **81** and a vacant space **82** particularly in vicinity of the attachment holes **17** can be deterred within the case **13**. Thus, even when tapping is performed on the attachment holes **17** with the tapping screws **18**, decay of threads threaded by the tapping due to an insufficient strength caused by the crack **81** or the vacant space **82** can be prevented. Therefore, occurrence of torque loss can be deterred, and free turn of the tapping screw **18** can be prevented. As a result, damaging the case **13** when the tapping screws **18** are fastened can be deterred.

(26) The attachment holes **17a**, **17b**, **17c** and **17d** are arranged in attachment hole seats **23a**, **23b**, **23c** and **23d**, respectively. Hereinafter, the attachment hole seats **23a**, **23b**, **23c** and **23d** are called “attachment hole seats **23**” when no particular distinction is required among them.

(27) Each of the attachment hole seats **23** has a projection part **30** that projects upward with respect to a case upper surface **29** around the attachment hole seat **23**. The projection part **30** has a peripheral dimension **d2** equal to or larger than 0.5 mm and equal to or smaller than 4.0 mm surrounding the attachment hole **17**. Preferably, the peripheral dimension **d2** is equal to or larger than 1.0 mm and equal to or smaller than 2.0 mm. A height dimension **d3** from the case upper surface **29** to the upper end of the projection part **30** is equal to or larger than 1.0 mm and equal to or smaller than 4.0 mm. Preferably, the height dimension **d3** is equal to or larger than 1.5 mm and equal to or smaller than 2.0 mm. The lower surface of the printed board **4** is in contact with the upper surface of the projection part **30** provided in the attachment hole seat **23** so that the printed board **4** is fixed to the case **13**.

(28) As shown in FIG. 4, a length dimension **d4** of the attachment hole **17** from the attachment hole seat **23**, that is, the upper end of the projection part **30** to the end part **19** is longer than a length dimension **d5** of the tapping screw **18**. A thickness dimension **d6** of the case **13** from the end part **19** of the attachment hole **17** to a lower end **13a** of the case **13** is equal to or larger than a predetermined first dimension. For example, the thickness dimension **d6** is equal to or larger than 1.0 mm and equal to or smaller than 6.00 mm. Preferably, the thickness dimension **d6** is equal to or larger than 1.5 mm and equal to or smaller than 4.0 mm. Preferably, the thickness dimension **d6** is equal to or larger than 2.0 mm and equal to or smaller than 3.0 mm.

(29) Similarly, as shown in FIG. 2, a thickness dimension **d7** of the case **13** from the attachment hole **17** to a side end **13b** of the case **13** is equal to or larger than a predetermined second dimension. For example, the thickness dimension **d7** is equal to or larger than 0.5 mm and equal to or smaller than 4.0 mm. Preferably, the thickness dimension **d7** is equal to or larger than 0.5 mm and equal to or smaller than 2.0 mm. More preferably, the thickness dimension **d7** is equal to or larger than 0.5 mm and equal to or smaller than 1.5 mm. Thus, retention of a resin in vicinity of the end part **19** of the attachment hole **17** during resin molding of the case **13** can be prevented, and occurrence of a crack **81** or a vacant space **82** in a detail within the case **13** can be deterred. Furthermore, the strength of the case **13** itself can be increased, and decay of the thread threaded by the tapping can be prevented. Therefore, occurrence of torque loss can be deterred, and free turn of the tapping screw **18** can be prevented. As a result, damaging the case **13** when the tapping screw **18** is fastened can be deterred.

(30) The case **13** has, at its outer circumferential corners, apparatus hole seats **25** for attaching the external apparatus **11**. The apparatus hole seats **25** have apparatus holes **24a**, **24b**, **24c** and **24d**. The apparatus holes **24a**, **24b**, **24c** and **24d** are used for attaching the case **13** to the external apparatus **11** by for example having bolts etc. (not illustrated) passing through themselves. Hereinafter, the apparatus holes **24a**, **24b**, **24c** and **24d** are called “apparatus holes **24**” when no particular distinction is required among them. Connection members such as bolts are screwed to the external apparatus **11** through the apparatus holes **24a**, **24b**, **24c** and **24d** so that the external apparatus **11** is attached to the lower surface of the semiconductor module **1**.

(31) As shown in FIG. 2, a gate G of a die is provided at a position corresponding to one end 27a of the case 13. Preferably, gates G of the die are provided at positions corresponding to both sides of the one end 27a of the case 13, that is, at positions corresponding to apparatus hole seats 25c and 25d. The case 13 is resin-molded with a resin material injected through the gates G of the die. In this case, another end 27b of the case 13 has the attachment holes 17a and 17b. The one end 27a of the case 13 has the attachment holes 17c and 17d. When a resin material flows during the resin molding, the resin tends to flow uniformly all around the attachment holes 17c and 17d close to the gates G. On the other hand, the resin flows while catching the air in areas close to the attachment holes 17a and 17b far away from the gates G, and a vacant space or a void tends to easily occur. Even when the attachment holes 17a and 17b are provided far away from the gates G, because the resin material flows smoothly along the rounded end parts 19 during resin molding, occurrence of a vacant space or a void within the case 13 can be deterred. Furthermore, the gates G of the die are provided at positions corresponding to both sides of the one end 27a of the case 13. Therefore, the resin material can flow smoothly toward the other end 27b linearly along the frame of the case 13, and occurrence of a vacant space or a void within the case 13 can be deterred.

(32) The apparatus holes 24a, 24b, 24c and 24d are disposed in apparatus hole seats 25a, 25b, 25c and 25d. Hereinafter, the apparatus hole seats 25a, 25b, 25c and 25d are called “apparatus hole seats 25” when no particular distinction is required among them. As shown in FIG. 2, a thickness dimension d8 of the case 13 from each of the apparatus hole seats 25 to the side end 13b of the case 13 is equal to the thickness dimension d7 of the case 13 from the corresponding attachment hole 17 to the side end 13b of the case 13. As shown in FIG. 4, the lower end (end part) 19 of the attachment hole 17 is provided at a deeper position than an upper surface of the apparatus hole seat 25. As shown in FIG. 2, the attachment holes 17 are disposed inner than the apparatus hole seats 25 in the length direction of the case 13 in planar view. In other words, the apparatus hole seats 25 are disposed more closely to the end parts (one end 27a and the other end 27b) than the attachment holes 17 in the length direction of the case 13 in planar view.

(33) A thickness dimension d9 of each of the apparatus hole seats 25 is larger than the thickness dimension d6 of the case 13 from the end part 19 of the attachment hole 17 to the lower end 13a of the case 13. The thickness dimension d9 of the apparatus hole seat 25 is equal to or larger than 2.0 mm and equal to or smaller than 8.0 mm. Preferably, the thickness dimension d9 is equal to or larger than 3.0 mm and equal to or smaller than 6.0 mm.

(34) A situation is now described in which a gate G of a die is provided at a position corresponding to the apparatus hole seat 25 at the one end 27a of the case 13. First of all, a situation where a gate G is provided at a position corresponding to the right side R of the case 13 in FIG. 4 and a resin material flows from the gate G, that is, a situation where the attachment hole 17 is close to the gate G is described. The resin material flowing from the gate G passes through the apparatus hole seat 25 where the flow path through which the resin material flows is narrow and collides with the end part 19 of the attachment hole 17 at a high speed. Here, if the end part 19 of the attachment hole 17 is not rounded, the resin material collides with the end part 19 at a high speed, resulting in a turbulent flow of the resin material and easily causing a cavity. On the other hand, according to this embodiment, because a resin material flows from the gate G smoothly along the rounded end part 19, occurrence of a vacant space or a void can be deterred.

(35) A situation where a gate G of a die is provided at a position corresponding to the left side L of the case 13 in FIG. 4 and a resin material flows from the gate G provided at the position corresponding to the left side L, that is, a situation where the attachment hole 17 is far away from the gate G is described. Because the attachment hole 17 is far away from the gate G, the viscosity of the resin increases while the resin material is flowing from the gate G to the attachment hole 17. Here, if the end part 19 of the attachment hole 17 is not rounded, the speed of the flow of the resin material further decreases in vicinity of the end part 19, and a vacant space or a void easily occurs in the subsequent apparatus hole seat 25. On the other hand, according to this embodiment, because

a resin material flows smoothly along the rounded end part **19**, the decrease of the resin speed can be suppressed, and occurrence of a vacant space or a void can be deterred.

(36) As shown in FIG. **4**, the inside diameter **d1** of attachment hole **17** on the end part **19** side is slightly smaller than a diameter **d10** of the tapping screw **18**. An inside diameter **d11** of an entrance **26** of the attachment hole **17** is wider than the inside diameter **d1** of the attachment hole **17** on the end part **19** side. Thus, when the tapping screw **18** is inserted to the attachment hole **17**, a worker can easily insert the tapping screw **18** to the attachment hole **17**. Therefore, the working efficiency for producing the semiconductor module **1** can be increased.

(37) As shown in FIG. **2**, the external terminals **14** at least include a P(+) input terminal **14a**, an N(-) input terminal **14b**, a UVW output terminal **14c**, a control terminal **14d** for the semiconductor module **1**, and an emitter terminal **14e** for temperature sensing within the case **13**. As shown in FIG. **2**, the P(+) input terminal **14a** is disposed at the one end **27a** of the case **13**, the N(-) input terminal **14b** is disposed at the other end **27b** of the case **13**, the UVW output terminal **14c** is disposed on the lower side of the case **13** in planar view, the control terminal **14d** is disposed on the upper side of the case **13** in planar view, and the emitter terminal **14e** is disposed at an upper right area of the case **13** in planar view. Each of the input terminal **14a**, input terminal **14b**, output terminal **14c**, control terminal **14d** and emitter terminal **14e** included in the external terminals **14** has one end projecting from the inner wall surface of the case **13** and another end projecting from the upper surface of the case **13**.

(38) It should be noted that the present invention is not limited to the aforementioned embodiment, but various changes can be made thereto. The sizes, shapes and functions of the constituent elements shown in the attached drawings according to the aforementioned embodiment are not limited thereto but can be changed as required without departing from the scope that the effect of the present invention is exerted. In addition, changes can be made thereto without departing from the scope of the object of the present invention.

(39) Having described that four attachment holes **17** are molded in the semiconductor module **1** according to the aforementioned embodiment, at least one attachment hole **17** may be molded, without limiting thereto. Although the apparatus holes **24** are resin-molded integrally with the case **13** in the semiconductor module **1** according to the aforementioned embodiment, the apparatus holes **24** may be molded in a separate body from the case **13**.

INDUSTRIAL APPLICABILITY

(40) As described above, the present invention has the effect that occurrence of a crack or a vacant space during resin molding of a case of a semiconductor module can be deterred and is particularly useful for a method for manufacturing a semiconductor module.

(41) The subject application is based on Japanese Patent Application No. 2019-144576 filed Aug. 6, 2019, the entirety of which is incorporated herein.

Claims

1. A semiconductor module, comprising: a semiconductor element; and a molded resin case having a sidewall portion surrounding the semiconductor element, the sidewall portion having an upper surface and a lower surface opposite to each other in a depth direction, a printed board being attachable to the upper surface, the sidewall portion having an attachment hole extending from the upper surface thereof in the depth direction and does not reach the lower surface thereof so as to have a tip portion at a bottom thereof within the case, a screw being insertable into the attachment hole to attach the printed board to the case, the attachment hole being integrally provided in the case, wherein the attachment hole has a columnar shape as a whole, and the tip portion has a hemispherical shape such that a resin material of the case is flowable along an outer surface of the tip portion, a shortest distance from an edge of the attachment hole to an end of the sidewall portion of the molded resin case in a plan view of the semiconductor module is equal to or less than a

prescribed value, and a thickness in the depth direction of the case from the bottom of the attachment hole to the lower surface of the case is in a range of 1.0 mm to 6.0 mm.

2. The semiconductor module according to claim 1, wherein the attachment hole is a screw hole for a tapping screw.
 3. The semiconductor module according to claim 2, wherein a length of the attachment hole is longer than a length of the tapping screw.
 4. The semiconductor module according to claim 1, wherein the case has a rectangular shape in a plan view of the semiconductor module, and the sidewall forms four sides of the case, one side of the four sides formed by the sidewall has a trace indicating that an injection of a resin material has been made, and the attachment hole is disposed at another side of the four sides formed by the sidewall that faces the one side.
 5. The semiconductor module according to claim 1, wherein the sidewall of the case has an apparatus hole seat having an apparatus hole for attaching an external apparatus, a bottom of the tip portion of the attachment hole is located at a position lower from the upper surface of the sidewall of the case in the depth direction than is a position of a surface of the apparatus hole seat that is parallel to the upper surface of the sidewall portion of the case.
 6. The semiconductor module according to claim 1, further comprising the printed board attached to the case through the attachment hole with a tapping screw.
 7. The semiconductor module according to claim 1, wherein the shortest distance from the edge of the attachment hole to the end of the sidewall portion of the molded resin case in the plan view is in a range of 0.5 mm to 4 mm.
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